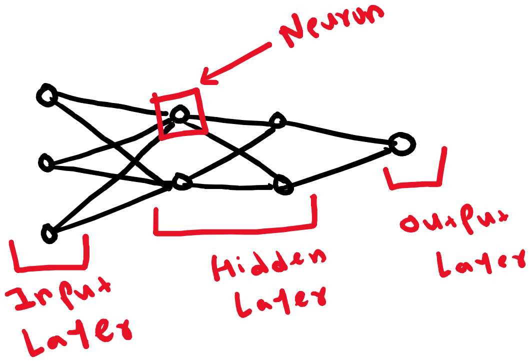
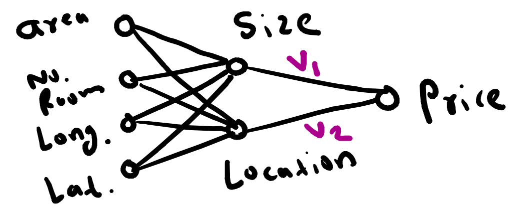
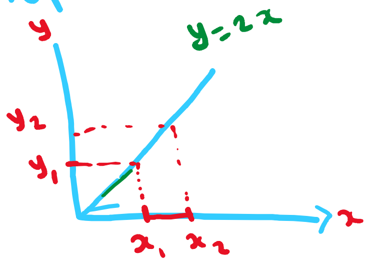




★ Derivatives: $\frac{dy}{dx}(x^2) = 2x$

* slope:



$x = 2, \quad y = 4$

$x = 2.001, \quad y = 4.002$

$\Delta x = 0.001, \quad \Delta y = 0.002$

$\text{slope} = \frac{0.002}{0.001} = 2$

$x = 3, \quad y = 6$

$x = 3.001, \quad y = 6.002$

$\Delta x = 0.001, \quad \Delta y = 0.002$

$\text{slope} = \frac{0.002}{0.001} = 2$

$\text{Slope} = \frac{\text{change in } y\ (\Delta y)}{\text{change in } x\ (\Delta x)} = \frac{y_2 - y_1}{x_2 - x_1}$